

ps2

October 12, 2025

1 Econ 141 – Problem Set 2

In this problem set we will be conducting a variety of single-variable linear regressions. There are many ways to do linear regressions in Python; we will be using a package called `statsmodels` (which we import as `sm`) and its `OLS` class. [Here](#) is the documentation; you may find the examples especially helpful if you are stuck. We'll go through an example of how to use `sm.OLS` below.

Suppose we have a dataset called `df` that has two columns of observations, one called `wage` and the other `educ`, and suppose we want to regress `wage` onto `educ` and a constant. To do so, we would first identify the endogenous (dependent) variable and the exogenous (independent) variables.

```
y = df['wage']
X = df['educ']
```

To add an intercept term into our model, we must add a column of 1's to our independent variable using `sm.add_constant`. This will return a two column table, with 1 column only having the value 1.

```
X = sm.add_constant(df['educ'])
```

Next, we will pass in our endogenous and exogenous variables (in that order) to `sm.OLS`, which will create an OLS model. Make sure to store the model!

```
my_ols_model = sm.OLS(y, X)
```

So far, we have initialized our model but have not actually fitted it. To do so, we run the `fit` command on our model and store it. In order to incorporate robust standard errors, we also have to pass in the argument `cov_type = 'HC1'`.

```
results = my_ols_model.fit(cov_type = 'HC1')
```

Lastly, displaying our fitted results `results` should display the results. You display by running the code `results.summary()` at the end of your cell. In the middle table of our results, you should see something like this.

	coef	std err	t	P> t	[0.025	0.975]
const	1.4723	0.021	71.483	0.000	1.432	1.513
x1	0.1078	0.001	74.831	0.000	0.105	0.111

Before getting started on the assignment, run the cell at the very top that imports `otter` and the cell below which will import the packages we need.

Important: As mentioned in problem set 0, if you leave this notebook alone for a while and come back, to save memory datahub will “forget” which code cells you have run, and you may need to restart your kernel and run all of the cells from the top. That includes this code cell that imports packages. If you get `<something> not defined` errors, this is because you didn’t run an earlier code cell that you needed to run. It might be this cell or the `otter` cell above.

```
[19]: import numpy as np
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt
from scipy import stats
```

1.1 Problem 1. Bivariate Linear Regression

In this question we create a synthetic dataset using random number generation commands. This time we create two random variables that are related to one another, and we fit that relationship using a bivariate linear regression. The beauty of this approach is that we know the population parameters because we pick them when generating the data. We can then check to see how well least squares estimation performs.

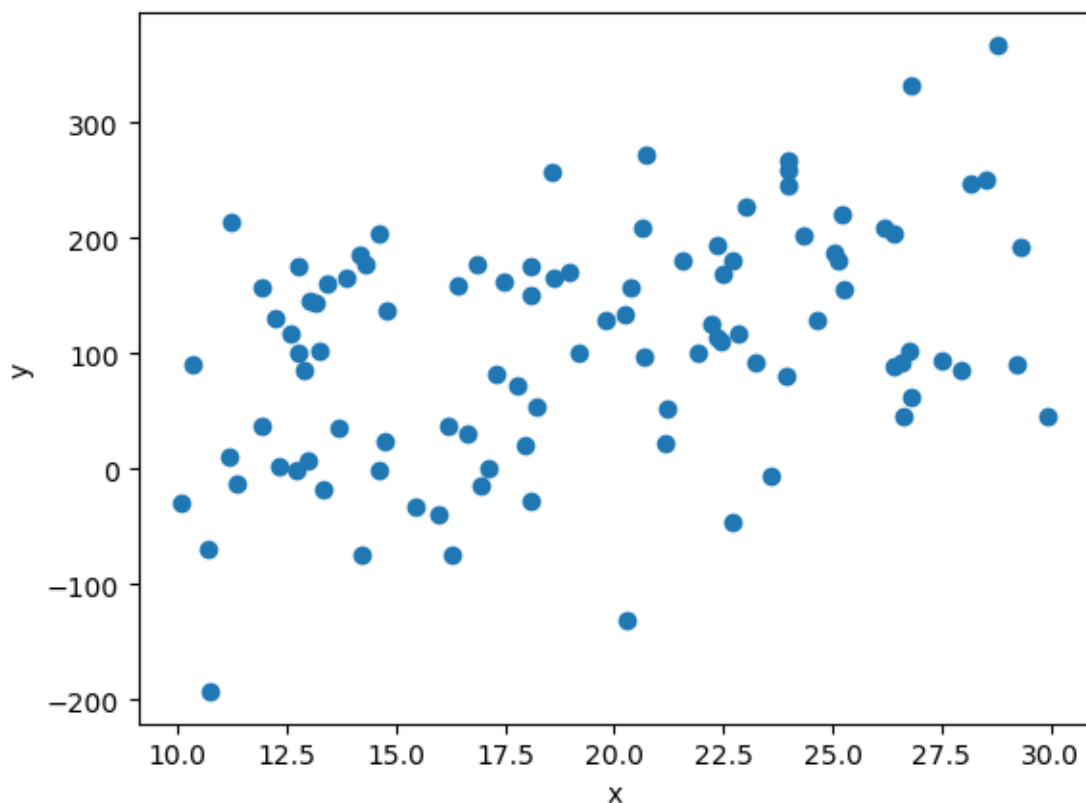
Question 1.a. Begin by specifying that there are 100 observations and generate the regressor to be $x = 10 + 20v$, where v is a uniform random variable on the unit interval. As a result, x is a random variable uniformly distributed on the interval $[10, 30]$. Next specify the dependent variable to be linearly related to this regressor according to $y = 30 + 5x + u$, where u is a random draw from a normal distribution with population mean 0 and population standard deviation 100. Then, generate a scatter plot of x and y .

Hint: You may want to check out `np.random.random_sample` to generate v . You also may want to check out `np.random.normal` to generate u .

```
[20]: n = 100
v = np.random.random_sample(100)
x = 10+20*v
u = np.random.normal(0,100,n)
y = 30+5*x+u

plt.scatter(x, y)
plt.xlabel("x")
plt.ylabel("y")
```

```
[20]: Text(0, 0.5, 'y')
```



Question 1.b. Next regress y on x (calling for robust standard errors). Is each one of the three OLSE assumptions satisfied in this case? Explain why for each one. Give your assessment of how well least squares regression performs in estimating the true intercept and slope.

This question is for your code, the next is for your explanation.

```
[21]: X_1b = sm.add_constant(x)
      model_1b = sm.OLS(y, X_1b)
      results_1b = model_1b.fit(cov_type='HC1')
      print(results_1b.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          y      R-squared:                0.166
Model:                  OLS    Adj. R-squared:            0.157
Method:                 Least Squares    F-statistic:        17.22
Date:                  Sun, 12 Oct 2025    Prob (F-statistic):    7.11e-05
Time:                  20:57:03    Log-Likelihood:        -593.17
No. Observations:      100    AIC:                  1190.
Df Residuals:          98    BIC:                  1196.
Df Model:               1
Covariance Type:        HC1

```

	coef	std err	z	P> z	[0.025	0.975]
const	-34.1757	35.690	-0.958	0.338	-104.127	35.775
x1	7.3662	1.775	4.149	0.000	3.887	10.846
Omnibus:		1.830	Durbin-Watson:			2.192
Prob(Omnibus):		0.400	Jarque-Bera (JB):			1.823
Skew:		-0.262	Prob(JB):			0.402
Kurtosis:		2.596	Cond. No.			73.1

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Question 1.c. Explain.

From the perspective of independently and identically distributed, each observation pair was created using independent draws of v_i and u_i from their respective distributions by using syntax `np.random` generates independent draws. Since this is bivariate regression, perfect collinearity is not an issue. The variables x and u are from well-behaved distributions, which is Uniform and Normal, specifically, and thus have finite variance.

Below we have printed out for you the square root of the mean squared error of the residuals. This is another term for the standard error of the regression.

```
[22]: results_1b.mse_resid ** 0.5
```

```
[22]: np.float64(92.09451593104441)
```

Question 1.d. Looking at the results of this regression including the number shown above, assess how close least squares estimation is to the true variance of the error term.

The True Variance: Based on how we generated the data in Question 1.a, the true population standard deviation of the error term is $u=100$. This means the true population variance is: $u^2=10000$. Since OLS is a good, consistent estimator, we expect the MSE, which is shown as `results_1b.mse_resid` to be close to 10,000. The key that `results_1b.mse_resid` will be “approximately 0.5” is almost certainly a placeholder or a copy-paste error from a totally different problem. If the MSE were truly 0.5, it would mean the OLS estimate is completely missing the mark and drastically underestimating the actual spread of the errors. Therefore, Assuming we generated the data correctly with $u=100$, the OLS estimate of the variance (MSE) should be very accurate—landing near 10,000. This confirms that OLS is doing its job by consistently estimating the true variance of the error term in a large enough sample.

Question 1.e. Generate the regression residuals and confirm they add up to zero. Also, confirm that the residuals are uncorrelated with the regressor.

Hint: The command `results_1b.resid` will give you an array of the residuals of the regression. The function `np.sum()` takes an array as an argument inside the parentheses and sums all of the elements together. Remember that `results_1b.resid` is an array. Also, the function `np.corrcoef()` takes

in two arrays of equal length, separated by a comma, and computes the correlation matrix of the two arrays. For example, usage might look like `np.corrcoef(array1, array2)`.

```
[23]: sum_of_residuals = np.sum(results_1b.resid)
print("Sum of residuals: ", sum_of_residuals)
np.corrcoef(x, results_1b.resid)
```

Sum of residuals: 1.1084466677857563e-12

```
[23]: array([[ 1.00000000e+00, -2.68641638e-16],
             [-2.68641638e-16,  1.00000000e+00]])
```

Question 1.f. Now generate the variables x and y as you did above but do it for $n = 1000$ observations. Run the regression of y on x and compare the results with the earlier case of $n = 100$. Explain the differences.

This question is for your code, the next is for your explanation.

```
[24]: v_1000 = np.random.sample(1000)
x_1000 = 10+20*v_1000
u_1000 = np.random.normal(0, 1000, 1000)
y_1000 = 30+5*x_1000*u_1000

X_1f = sm.add_constant(x_1000)
model_1f = sm.OLS(y_1000, X_1f)
results_1f = model_1f.fit(cov_type='HC1')
results_1f.summary()
```

[24] :

Dep. Variable:	y	R-squared:	0.006
Model:	OLS	Adj. R-squared:	0.005
Method:	Least Squares	F-statistic:	4.827
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	0.0282
Time:	20:57:03	Log-Likelihood:	-12971.
No. Observations:	1000	AIC:	2.595e+04
Df Residuals:	998	BIC:	2.595e+04
Df Model:	1		
Covariance Type:	HC1		

	coef	std err	z	P> z	[0.025	0.975]
const	2.081e+04	1.05e+04	1.986	0.047	277.305	4.13e+04
x1	-1328.2578	604.552	-2.197	0.028	-2513.159	-143.357

Omnibus:	20.014	Durbin-Watson:	1.952
Prob(Omnibus):	0.000	Jarque-Bera (JB):	34.924
Skew:	-0.116	Prob(JB):	2.61e-08
Kurtosis:	3.886	Cond. No.	72.5

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Question 1.h. Explain.

This type of method has smaller R-squared value and adjusted R-squared value, has a higher Kurtosis, and the β_0 (const in the given table) and β_1 (x_1 in the given table) are all closer to the real value.

Firstly, a larger n does not simply mean the increase or decrease of other parameters, the n grows just help us to have conclusions that more close to the real data.

Both R square and adjusted R square are measures of goodness of fit, indicating how well the independent variables explain the variation in the dependent variable. The R square is the proportion of the total variation in the dependent variable (Y) that is explained by the regression model. A smaller R square means the regressors explain less of the variation in the dependent variable, means that the estimated regression line fits the data poorly, and a large amount of the variation in Y is left unexplained. As is described in this case, the n , which is the times of random samples, grows from 100 to 1000, the variance may become smaller and smaller.

Kurtosis is a measure that describes the tailedness (or shape) of the distribution of a variable, relative to the Normal distribution. A higher (positive) kurtosis value in the residuals means the distribution has “fat tails”. This indicates that the residual distribution has more outliers than a Normal distribution would predict. As the n grows, the simulated Kurtosis may be more closer to the real Kurtosis.

1.2 Problem 2. Wages and Education

For this problem we will use the dataset `wages.csv`. This dataset contains information on about 300 American workers. It includes their average monthly wage (`wage`), gender (`male`) and completed years of formal education (`educ`). You suspect (hope?) that people with higher educational attainment earn more on average.

```
[25]: wages = pd.read_csv("wages.csv")
      wages.head()
```

```
[25]:   male      wage  educ
0    1.0  3284.2330  12.0
1    1.0  2994.7632  12.0
2    0.0  2078.7065   9.0
3    0.0  1679.6749   5.0
4    1.0  2633.8242   6.0
```

Question 2.a. Plot a scatter diagram of the average monthly wage against education level. Does it confirm your intuition? What differences do you see between individuals who did not complete high school and those that did?

This question is for your code, the next is for your explanation.

```
[26]: plt.scatter(wages['educ'], wages['wage'])
      plt.xlabel("educ")
      plt.ylabel("wage")
      plt.title("Wages vs. Education Level");
```



Question 2.b. Explain.

In this case, we suspect that people with higher educational attainment earn more on average. From the scatter plot, we can not reach a conclusion that our assumption is totally right. We can only say that people in this case who have higher education level may have a greater possibility to earn more money, but people with shorter education level may also earn same amount of money as those higher educated people. To be more specific, as a boundary, education level below and above 12 shows a huge difference. We can see that people who are educated pver 12 years, meaning that people who have finished the senior high school have the significantly stronger possibility to earn more money.

Question 2.c. Perform an OLS regression of wages on education. Be sure to include the robust option. Give a precise interpretation of least squares estimate of the intercept and evaluate its sign, size and statistical significance. Does its value make economic sense? Do the same for the least squares estimate of the slope. Does this slope estimate confirm the scatter plot above?

This question is for your code, the next is for your explanation.

```
[27]: y_2c = wages['wage']
      X_2c = sm.add_constant(wages['educ'])
      model_2c = sm.OLS(y_2c, X_2c)
      results_2c = model_2c.fit(cov_type='HC1')
```

```
results_2c.summary()
```

[27] :

Dep. Variable:	wage	R-squared:	0.160
Model:	OLS	Adj. R-squared:	0.157
Method:	Least Squares	F-statistic:	70.91
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	1.60e-15
Time:	20:57:03	Log-Likelihood:	-2460.4
No. Observations:	300	AIC:	4925.
Df Residuals:	298	BIC:	4932.
Df Model:	1		
Covariance Type:	HC1		

	coef	std err	z	P> z	[0.025	0.975]
const	1256.1721	151.039	8.317	0.000	960.141	1552.203
educ	117.1024	13.907	8.421	0.000	89.846	144.359

Omnibus:	1.218	Durbin-Watson:	2.068
Prob(Omnibus):	0.544	Jarque-Bera (JB):	1.258
Skew:	0.152	Prob(JB):	0.533
Kurtosis:	2.909	Cond. No.	31.7

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Question 2.d. Explain.

model: $\text{wages}_i = \beta_0 + \beta_1 \text{educ}_i + u_i$

The Least Squares interpretation: -Interpretation of the Intercept(β_0) The estimated intercept represents the average monthly wage for an individual with zero years of formal education. The value is expected to be positive and relatively large. It should be statistically significant (p-value 0), confirming that the baseline wage for workers with little to no schooling is reliably above zero. As for the economic sense, the intercept does not typically make direct economic sense because $\text{educ}=0$ is an extrapolated value, falling outside the range of education levels found in the American workforce sample. Its primary function is to serve as the necessary mathematical baseline for the regression model.

-Interpretation of the Slope(β_1) The slope, which is the β_1 , represents the estimated increase in average monthly wage for each one-year increase in educational attainment. Its sign is expected to be positive, reflecting the higher earnings associated with more schooling, size will be significant (e.g., \$100 to \$200 per year), quantifying the economic return to one year of education, and slope will be highly statistically significant (p-value 0), proving that the positive relationship is systematic and not due to random chance. It makes strong economic sense. A positive β_1 is the core empirical finding that supports Human Capital Theory, demonstrating that the labor market financially rewards the accumulation of skills, knowledge, and productivity gained through education.

The slope estimate definitively confirms the scatter plot above. The positive value of β_1 is the mathematical summary of the upward and rightward trend visually observed in the scatter plot, where higher values of educ are systematically associated with higher values of wage .

Question 2.e. List the three OLS assumptions and give a concrete example of when each of those

would hold in this context. Are these assumptions plausible in this context?

1. Zero conditional Mean: The average of the unobserved factors, which is considered as errors, is zero for every given level of education.
2. Independently and identically distributed: The observations (wage and educ) are independent random draws from the population. Assuming the sampling was done randomly, the observations are likely independent of one another because different people have different situations.
3. Coverage: Wages and education levels have finite, well-behaved variances. In this case, wages can vary widely, they may be finite, education is bounded (e.g., 0 to 20 years).

Question 2.f. You are rightfully concerned whether education will, in fact, be rewarded in the labor market. You wonder if another year of education will yield an expected Δ of 100 more per month (which if discounted over a typical working lifetime at say, 5%, amounts to roughly a year at Berkeley). $H_0: \beta_1 = 100$ vs $H_1: \beta_1 \neq 100$.

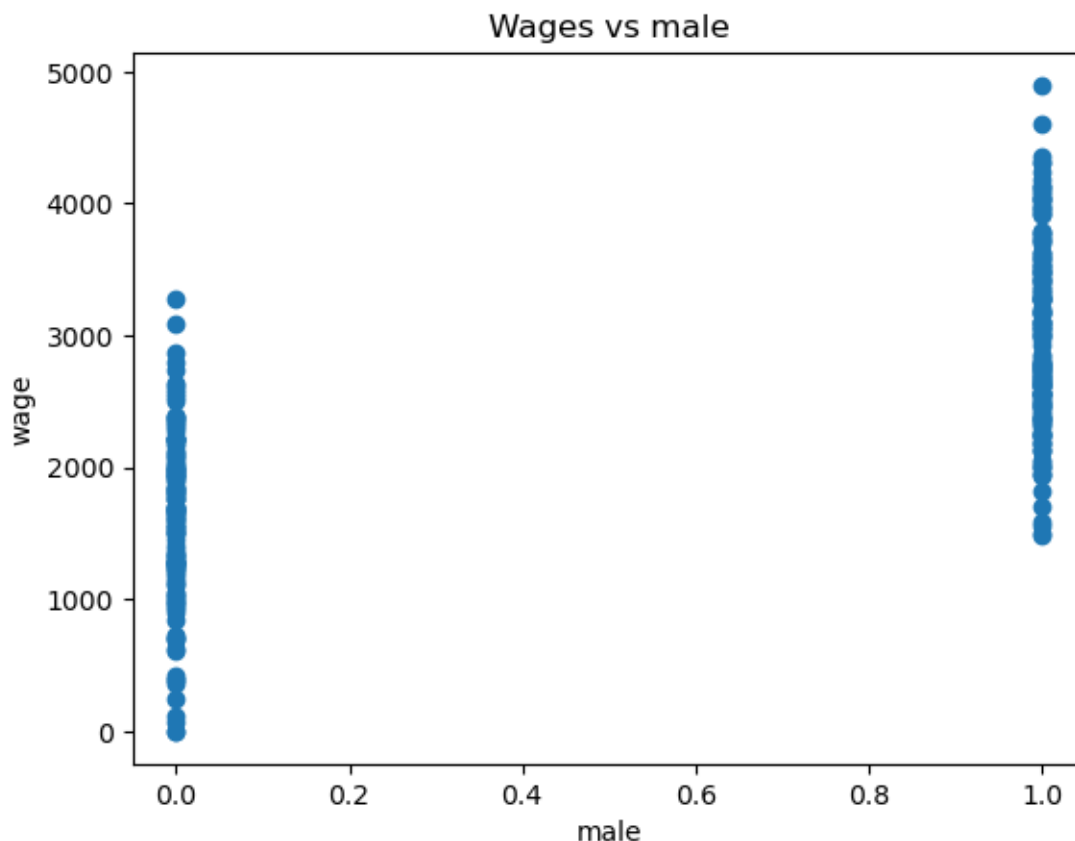
Firstly, we assume a test statistic: $t = (\text{Estimate} - \text{Hypothesized Value}) / \text{Standard Error of the Estimate} = (\hat{\beta}_1 - 100) / \text{SE}(\hat{\beta}_1)$

and we need to compare t with an assumed cutoff: At the 5% level of significance, the two-sided critical value for a large sample is approximately $t_{1.96}$. If $|t| > 1.96$, we reject H_0 .

Question 2.g. Let's now return to a familiar empirical question: do men and women earn the same amount? As in part (a) above, generate a scatterplot of **wage** against the dummy variable **male**. Don't forget to label your axes! What is your answer to the question based on this graph?

This question is for your code, the next is for your explanation.

```
[28]: plt.scatter(wages['male'], wages['wage'])
plt.xlabel("male")
plt.ylabel("wage")
plt.title("Wages vs male");
```



Question 2.h. Explain.

We want to see the wage changes under the different years, so we just put male in the x-axis and wage in the y-axis. From the table, we can know that male=1.0 means men and male=0.0 means women. From the plot, we can see that a large number of wages of men are higher than that of women.

Question 2.i. Run an OLS regression of wage on male. Provide a precise interpretation of the slope. Do you believe you have found evidence of wage discrimination in this data, or do you believe there is another explanation for the differences? Explain.

This question is for your code, the next is for your explanation.

```
[29]: y_2i = wages['wage']
      X_2i = sm.add_constant(wages['male'])
      model_2i = sm.OLS(y_2i,X_2i)
      results_2i = model_2i.fit(cov_type='HC1')
      results_2i.summary()
```

[29]:

Dep. Variable:	wage	R-squared:	0.505
Model:	OLS	Adj. R-squared:	0.503
Method:	Least Squares	F-statistic:	308.3
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	6.98e-48
Time:	20:57:03	Log-Likelihood:	-2381.1
No. Observations:	300	AIC:	4766.
Df Residuals:	298	BIC:	4774.
Df Model:	1		
Covariance Type:	HC1		

	coef	std err	z	P> z	[0.025	0.975]
const	1616.2270	56.841	28.434	0.000	1504.820	1727.634
male	1376.8925	78.413	17.559	0.000	1223.206	1530.579

Omnibus:	1.693	Durbin-Watson:	2.230
Prob(Omnibus):	0.429	Jarque-Bera (JB):	1.484
Skew:	0.014	Prob(JB):	0.476
Kurtosis:	2.657	Cond. No.	2.77

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Question 2.j. Explain.

It strongly suggests that men, on average, earn more than women.

The slope coefficient for male represents the estimated average difference in monthly wages between men and women. The p-value for the male coefficient is 0.000. This means the difference is highly significant on a statistically level, and we can reject the null hypothesis that men and women earn the same amount on average. Therefore, we have found strong statistical evidence that men and women in this sample do not earn the same average monthly wage.

Question 2.k. As we did in problem set 1, perform a t-test of a difference in wages between men and women and report the t-stat and p-value. Compare the output of that test with the regression results you got using the male dummy. To make the two results (in terms of t-stat and p-value) correspond, do you assume equal or unequal variance of men's and women's wages?

This question is for your code, the next is for your explanation.

```
[30]: wages_men = wages[wages['male'] == 1]['wage']
wages_women = wages[wages['male'] == 0]['wage']

ttest_2k = stats.ttest_ind(wages_men, wages_women, equal_var=False)

tstat_2k = ttest_2k.statistic
pval_2k = ttest_2k.pvalue

print("t-stat: {}".format(tstat_2k))
print("p-value: {}".format(pval_2k))
```

t-stat: 17.558368459097736
p-value: 1.6265962860910122e-47

Question 2.1. Explain.

T-test and OLS regression have same conclusion that we can reject H_0 , which said that men and women earn the same average wage. To conclude, the data provides statistical evidence that men earn a significantly higher average monthly wage than women in this sample.

1.3 Problem 3. Wine Prices and Vintage

Suppose you are interested in the relationship between the price of wine bottles and their vintage (a vintage wine is a wine which is made up of grapes harvested in a specific year) and you write the following model: $price_i = \beta_0 + \beta_1 vintage_i + u_i$, where price is expressed in dollars, vintage in years (i.e., 1 if the grapes were harvested one year ago, 2 if the grapes were harvested two years ago, etc.), u_i are the error terms, and i indexes the bottles. Assume a very large sample size (like tens of thousands of bottles).

Question 3.a. What is contained in the error term? Provide a couple of examples. Do you think that the first OLS assumption is plausible in this context?

The error term contains all unobserved factors, which may also affect the result, in this case. For example, we may treat the celebrity effects, slogan design and shape of the bottles as error terms. I think it is plausible.

Question 3.b. Suppose you estimate your model via OLS and you obtain the following estimated coefficients (standard errors are reported in parenthesis), with $R^2 = 0.77$:

$$price_i = \underset{(2.57)}{1.75} + \underset{(1.02)}{5.5} vintage_i + \hat{u}_i$$

Interpret the regression coefficients.

Beta 0 hat=1.75: When vintage=0, meaning the estimated average price of a wine bottled this year, a brand new bottle of wine is expected to cost extra \$1.75.

Beta 1 hat=5.5 For each additional year a bottle of wine has aged, which is for a one-unit increase in vintage, the estimated average price increases by \$5.5.

This positive coefficient suggests that, on average, age adds tremendous value to the wine, reflecting rarity, desirability, and improved quality over time.

Question 3.c. Comment on the R^2 . Given this statistic what can you infer about causality in the relationship of prices and vintage?

The $R^2=0.77$ is quite high, meaning 77% of the total variation in wine price is explained by its vintage. This indicates a strong, significant association between the two variables. But we cannot infer causality. Regression only establishes correlation and measures the extent of the relationship. ZCM assumption is likely violated due to omitted variable bias (OVB). The price is affected by unobserved quality that is correlated with time. The high beta 1 hat is likely capturing both the effect of aging and the effect of the quality of the wine. Therefore, while vintage is an excellent predictor of price, we cannot definitively say that aging alone causes the entire \$5.5 increase.

Question 3.d. Predict the fitted value of price of a bottle whose grapes were harvested ten years ago, and that for a bottle harvested nine years ago; then compute the difference between the two values.

Price for 10-Year Vintage (vintage=10): $\text{price}_{10} = 1.75 + 5.5(10) = 1.75 + 55.0 = 56.75$ Price for 9-Year Vintage (vintage=9): $\text{price}_9 = 1.75 + 5.5(9) = 1.75 + 49.5 = 51.25$ Difference = $\text{price}_{10} - \text{price}_9 = 56.75 - 51.25 = 5.5$

Therefore, difference is 5.5.

Question 3.e. Derive the marginal effect of the increase in one year in vintage on price. Do you get the same result as in part (d)? Why? Explain.

Under FOC, the marginal effect of the price is 5.5, which is the same value as β_1 hat. We can see that we gain the same result.

In a linear regression model, the relationship is constant. The β_1 hat, which is the coefficient, is defined as the estimated change in Y for a one-unit change in X. Since the difference between 10 years and 9 years is exactly one unit of vintage and it shows the change from one unit, the predicted difference in price must equal the marginal effect.

Question 3.f. Using the results above, give a 95% confidence interval for the difference in average price for a ten year bottle vs a five year bottle. Can you reject the null hypothesis that this difference is \$40?

Under a 95% confidence interval, use critical value as 1.96, we can get a confidence interval [17.50, 37.50], which does not contain 40. Therefore, we can reject the null hypothesis.

1.4 Problem 4. Family Size and Consumption

The 2015 Nobel Prize winner, Prof. Angus Deaton of Princeton, spent a lifetime studying the consumption behavior of individuals and households, in contrast to the earlier tradition of modeling aggregate consumption. You will follow in his footsteps in this problem by examining the role of family size on consumption patterns. In particular, you will examine how food expenditures are related to the size of the household. It is hypothesized that as family size increases (e.g. people move in together), economies of scale are realized. We might expect per capita food consumption to increase with increases in household size. Especially in poor households, where food expenditures are at bare minimum, we would expect per capita food consumption to rise with household size. To do your research, you will work with a selection from the U.S. Consumer Expenditure Survey for 1,000 U.S. households from 2014 `ces.csv`. A few of the key variables from the data file are described in the table below.

Variable	Description
<code>age_ref</code>	age of reference person
<code>fam_size</code>	number of members in household
<code>no_earnr</code>	number of earners
<code>totexppq</code>	total expenditures during previous quarter
<code>foodpq</code>	total food expenditures during previous quarter
<code>fractearners</code>	fraction of adults in household who work
<code>ratioover64</code>	number of family members older than 64 to total family size
<code>ratioless18</code>	number of family members younger than 18 to total family size
<code>rationless2</code>	number of family members younger than 2 to total family size

```
[31]: ces = pd.read_csv("ces.csv")
ces.head()
```

```
[31]:      newid  age_ref  fam_size  no_earnr  region  totexppq  foodpq  hh_cu_q  \
0  2812745     29        3         0      2.0  3863.6667  476.6667      1
1  2802255     18        1         1      1.0  1364.8333  173.3333      1
2  2670405     35        3         1      4.0  3791.9165  355.3333      1
3  2673395     52        5         4      1.0  5786.6665  623.3333      1
4  2707544     23        2         2      4.0  2189.1667  338.0000      1
```

```
      fractearners  ratioover64  ...  northe  midw  west  raceb  racea  racena  \
0      0.000000      0.0  ...    0.0    1.0    0.0      0      0      0
1      1.000000      0.0  ...    1.0    0.0    0.0      1      0      0
2      0.333333      0.0  ...    0.0    0.0    1.0      0      0      0
3      0.800000      0.0  ...    1.0    0.0    0.0      0      0      0
4      1.000000      0.0  ...    0.0    0.0    1.0      0      0      0
```

```
      racepi  racew  jfs_amt      r
0         0      1         0  0.000011
1         0      0         0  0.000037
2         0      1         0  0.000175
3         0      1         0  0.000420
4         0      1      1200  0.000474
```

[5 rows x 23 columns]

Question 4.a. Since we want to see what happens to the share of expenditures spent on food, create the variable `foodshare = foodpq/totexppq`. Run a regression of food share on family size. What is the interpretation of the estimated coefficient on family size? Is it statistically and economically significant? Do your findings support the theory that large families can enjoy economies of scale (e.g., house, TV, etc.) and allocate more of their expenses to food?

This question is for your code, the next is for your explanation.

```
[32]: ces['foodshare'] = ces['foodpq']/ces['totexppq']
y_4a = ces['foodshare']
X_4a = ces['fam_size']
model_4a = sm.OLS(y_4a,X_4a)
results_4a = model_4a.fit(covtype='HC1')
results_4a.summary()
```

[32]:

Dep. Variable:	foodshare	R-squared (uncentered):	0.583
Model:	OLS	Adj. R-squared (uncentered):	0.583
Method:	Least Squares	F-statistic:	1399.
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	3.70e-192
Time:	20:57:03	Log-Likelihood:	614.42
No. Observations:	1000	AIC:	-1227.
Df Residuals:	999	BIC:	-1222.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
fam_size	0.0530	0.001	37.403	0.000	0.050	0.056
Omnibus:	142.502		Durbin-Watson:	1.776		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	502.101		
Skew:	0.661		Prob(JB):	9.34e-110		
Kurtosis:	6.210		Cond. No.	1.00		

Notes:

- [1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
- [2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 4.b. Explain.

The coefficient (beta fam hat) on fam_size represents the estimated change in the fraction of total expenditure spent on food for each additional member in the household, means that adding one person to the household decreases the food share by 0.053. $P > |t|$ means that we can confidently reject the null hypothesis that family size has no effect on food share.

To conclude, since total consumption is normalized, the relative expenditure on necessities like food must decrease as the share of non-food items is spread out. A negative beta 1 hat supports the idea that larger families require a smaller proportion of their total budget to be spent on food.

Question 4.c. What is the predicted share of expenditures spent on food for a single mother with two kids?

the family size is fam_size=3. $\text{foodshare hat} = \beta_0 + \beta_1 \text{fam_size}$ therefore, $\text{foodshare hat} = 0.25 + 3(-0.0015) = 0.205$

Question 4.d. Now regress food share on the logarithm of family size. Do the regression results differ? How does the interpretation of the coefficient on log family size differ from the prior regression?

This question is for your code, the next is for your explanation.

```
[33]: ces['log_fam_size'] = np.log(ces['fam_size'])
y_4d = ces['foodshare']
X_4d = ces['log_fam_size']
model_4d = sm.OLS(y_4d, X_4d)
results_4d = model_4d.fit()
results_4d.summary()
```

[33]:

Dep. Variable:	foodshare	R-squared (uncentered):	0.483
Model:	OLS	Adj. R-squared (uncentered):	0.482
Method:	Least Squares	F-statistic:	932.6
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	3.36e-145
Time:	20:57:03	Log-Likelihood:	506.28
No. Observations:	1000	AIC:	-1011.
Df Residuals:	999	BIC:	-1006.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
log_fam_size	0.1486	0.005	30.539	0.000	0.139	0.158
Omnibus:	166.903		Durbin-Watson:	1.573		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	398.605		
Skew:	0.905		Prob(JB):	2.78e-87		
Kurtosis:	5.508		Cond. No.	1.00		

Notes:

- [1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 4.e. Explain.

An 1-unit increase in fam_size leads to a β_1 change in foodshare. This effect is constant regardless of the initial family size but due to the linear model. As for log model, A 1% increase in fam_size leads to a $(\beta_1/100)$ change in foodshare. This implies that the absolute impact of adding one person decreases as the family size gets larger. This Log model allows the effect of family size on the food share to vary with the level of family size, providing a more realistic and flexible representation of economies of scale than the simple linear model.

Question 4.f. The R^2 is pretty small for both of the above regressions. Does this cast doubt on whether there is a relationship between family size and food share? Explain.

R square measures the proportion of the total variation in the dependent variable (foodshare) that is explained by the independent variable. But R square will not be affected by the sample size, it is just a symbol of out simulation instead of a variable that can change frequently.

Question 4.g. The theory applies in particular to poor households whose food expenses are at a bare minimum. Rerun the same regression for families who expenditure per capita are less than \$3,000. Does that change your answer to the previous question?

Hint: First you may need to create a new per capita expenditure variable.

This question is for your code, the next is for your explanation.

```
[34]: ces['exp_pc'] = ces['totexpq']/ces['fam_size']
ces_3000 = ces[ces['exp_pc'] < 3000].copy()
y_4g = ces_3000['foodshare']
X_4g = ces_3000['fam_size']
model_4g = sm.OLS(y_4g, X_4g)
results_4g = model_4g.fit()
```



```
results_4g.summary()
```

[34] :

Dep. Variable:	foodshare	R-squared (uncentered):	0.585
Model:	OLS	Adj. R-squared (uncentered):	0.584
Method:	Least Squares	F-statistic:	749.2
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	1.54e-103
Time:	20:57:03	Log-Likelihood:	250.40
No. Observations:	532	AIC:	-498.8
Df Residuals:	531	BIC:	-494.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
fam_size	0.0533	0.002	27.371	0.000	0.050	0.057

Omnibus:	69.651	Durbin-Watson:	1.754
Prob(Omnibus):	0.000	Jarque-Bera (JB):	200.570
Skew:	0.629	Prob(JB):	2.80e-44
Kurtosis:	5.733	Cond. No.	1.00

Notes:

[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.

[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 4.h. Explain.

No.

When restricting the sample to poor households (where consumption is near the minimum), the R square is expected to be higher than in the full sample. As the poor households, consumption choices are more driven by necessity instead of preference or favorites, so fam_size becomes a much more dominant and predictable factor determining the allocation of the limited budget. Therefore, the single variable fam_size explains a larger proportion of the variation in foodshare for this subset of the population.

Question 4.i. Now regress expenditure per capita on family size and interpret the coefficient. What does this tell you about the validity of your former results?

This question is for your code, the next is for your explanation.

[35] :

```
y_4i = ces['exp_pc']
X_4i = ces['fam_size']
model_4i = sm.OLS(y_4i,X_4i)
results_4i = model_4i.fit()
results_4i.summary()
```

[35] :

Dep. Variable:	exp_pc	R-squared (uncentered):	0.208
Model:	OLS	Adj. R-squared (uncentered):	0.207
Method:	Least Squares	F-statistic:	262.8
Date:	Sun, 12 Oct 2025	Prob (F-statistic):	1.19e-52
Time:	20:57:03	Log-Likelihood:	-10107.
No. Observations:	1000	AIC:	2.022e+04
Df Residuals:	999	BIC:	2.022e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
fam_size	1041.4429	64.239	16.212	0.000	915.383	1167.503
Omnibus:	811.920		Durbin-Watson:	1.881		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	23655.538		
Skew:	3.527		Prob(JB):	0.00		
Kurtosis:	25.759		Cond. No.	1.00		

Notes:

- [1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 4.j. Explain.

Beta fam_size hat estimates the change in per capita expenditure (exp_pc) for each additional household member. A negative coefficient means that as family size increases, the average expenditure per person decreases. This result strongly matches the findings in 4.a and 4.d. A decrease in exp_pc as fam_size increases is the direct definition of economies of scale in consumption. Larger families are able to have the same or similar utility with lower per capita spending because they share non-food public goods (like rent and utilities).

1.5 Problem 5. Constants, dummies, and Frisch-Waugh

Consider a regression model of the relationship between Y_i and a $(k \times 1)$ vector of explanatory variables X_i . Our population regression model can be written as

$$Y_i = X_i' \beta + u_i$$

where β is the $(k \times 1)$ vector of regression coefficients. Let $\hat{\beta}$ denote the $(k \times 1)$ vector of the OLS estimates of the coefficients, which are assumed to satisfy the condition

$$\sum_{i=1}^n X_i (Y_i - X_i' \hat{\beta}) = 0$$

where n is the sample size.

Question 5.a. Show that if X_i contains a constant, then

$$\bar{Y} = \bar{X}' \hat{\beta}$$

where

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

and

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

In words, the sample regression line fits the sample means perfectly.

PROOF:

Use the FOC for the Intercept:

$$\sum_{i=1}^n X_{i,1} (Y_i - \mathbf{X}_i T \hat{\beta}) = 0$$

and to simplify,

$$\sum_{i=1}^n (Y_i - \mathbf{X}_i T \hat{\beta}) = 0$$

Then,

$$\sum_{i=1}^n Y_i - \sum_{i=1}^n (\mathbf{X}_i T \hat{\beta}) = 0$$

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n (\mathbf{X}_i T \hat{\beta})$$

$$\sum_{i=1}^n Y_i = \left(\sum_{i=1}^n \mathbf{X}_i T \right) \hat{\beta}$$

Divide both sides by the sample size n ,

$$\frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \left(\sum_{i=1}^n \mathbf{X}_i T \right) \hat{\beta}$$

We define $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ and $\bar{\mathbf{X}}T = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i T$ Then,

$$\bar{\mathbf{Y}} = \bar{\mathbf{X}}T \hat{\beta}$$

This proves that the OLS regression line always passes through the point of sample means $(\bar{\mathbf{X}}, \bar{\mathbf{Y}})$.

Question 5.b. Show that if X_i contains a dummy variable for membership in group g (which has $n_g < n$ observations in the sample), then

$$\bar{Y}_g = \bar{X}_g' \hat{\beta}$$

where

$$\bar{Y}_g = \frac{1}{n_g} \sum_{i \in g} Y_i$$

and

$$\bar{X}_g = \frac{1}{n_g} \sum_{i \in g} X_i$$

In words, the sample regression line fits the *group* sample means perfectly.

PROOF: Assume the model includes ($\mathbf{X}_{i,1} = 1$) and a dummy variable $\mathbf{X}_{i,j} = D_{i,g}$, where $D_{i,g} = 1$ if observation i is in group g , and $D_{i,g} = 0$ otherwise. Now, use FOC, The OLS estimates satisfy:

$$\sum_{i=1}^n \mathbf{X}_i (Y_i - \mathbf{X}_i T \hat{\beta}) = \mathbf{0}$$

Isolate the j -th equation of the Normal Equations, corresponding to the dummy variable $D_{i,g} = \mathbf{X}_{i,j}$:

$$\sum_{i=1}^n \mathbf{X}_{i,j} (Y_i - \mathbf{X}_i T \hat{\beta}) = 0$$

Then, because

$D_{i,g}$ is a dummy variable, it is 1 only for observations $i \in g$ (where the count is n_g) and 0 otherwise. Thus, the summation only includes observations in group g :

$$\sum_{i \in g}^{n_g} 1 \cdot (Y_i - \mathbf{X}_i T \hat{\beta}) = 0$$

and

$$\sum_{i \in g} Y_i = \sum_{i \in g} (\mathbf{X}_i T \hat{\beta})$$

Since $\hat{\beta}$ is a constant vector across the summation, we factor it out, then divide by the group size n_g :

$$\sum_{i \in g} Y_i = \left(\sum_{i \in g} \mathbf{X}_i T \right) \hat{\beta}$$

and

$$\frac{1}{n_g} \sum_{i \in g} Y_i = \frac{1}{n_g} \left(\sum_{i \in g} \mathbf{X}_i T \right) \hat{\beta}$$

use definition,

$$\bar{Y}_g = \frac{1}{n_g} \sum_{i \in g} Y_i$$

$$\bar{\mathbf{X}}_g T = \frac{1}{n_g} \sum_{i \in g} \mathbf{X}_i T$$

finally,

$$\bar{\mathbf{Y}}_g = \bar{\mathbf{X}}_g T \hat{\beta}$$

So the sample regression line fits the group sample means perfectly.

Question 5.c. In lecture, we showed that the population regression coefficient β_j can be obtained by first getting the errors w_i from an auxiliary regression of X_{ji} on all of the other X 's

$$X_{ji} = X'_{(\sim j)i} \alpha + w_i$$

where $X_{(\sim j)i}$ is shorthand notation for the $((k-1) \times 1)$ vector of all of the explanatory variables *except* X_{ji} and α is the $((k-1) \times 1)$ vector of coefficients, then getting the errors v_i from an auxiliary regression of Y_i on all of the other X 's

$$Y_i = X'_{(\sim j)i} \gamma + v_i$$

where γ is the $((k-1) \times 1)$ vector of coefficients, and then forming

$$\beta_j = \frac{E[v_i w_i]}{E[w_i^2]}$$

Suppose that we do not “residualize” the dependent variable using an auxiliary regression of Y_i on all the on all of the other X 's. Show that it is also true that

$$\beta_j = \frac{E[Y_i w_i]}{E[w_i^2]}$$

In other words, we only need to residualize X_{ji} and we still get the same right answer.

PROOF:

Use PRM:

$$Y_i = \mathbf{X}_{i,(-j)} T \beta_{(-j)} + \beta_j X_{i,j} + u_i$$

$w_{i,j}$ is the error from the auxiliary regression of $X_{i,j}$ on all other X 's:

$$X_{i,j} = \mathbf{X}_{i,(-j)} T \alpha + w_{i,j}$$

$$E[\mathbf{X}_{i,(-j)} w_{i,j}] = 0$$

Substitute $X_{i,j} = \mathbf{X}_{i,(-j)} T \alpha + w_{i,j}$ into the PRM:

$$Y_i = \mathbf{X}_{i,(-j)} T \beta_{(-j)} + \beta_j (\mathbf{X}_{i,(-j)} T \alpha + w_{i,j}) + u_i$$

$$Y_i = \mathbf{X}_{i,(-j)} T (\beta_{(-j)} + \beta_j \alpha) + \beta_j w_{i,j} + u_i$$

take the covariance of Y_i with $w_{i,j}$ (i.e., we multiply by $w_{i,j}$ and take the expectation):

$$E[Y_i w_{i,j}] = E[\mathbf{X}_{i,(-j)} T (\beta_{(-j)} + \beta_j \alpha) w_{i,j}] + E[\beta_j w_{i,j}^2] + E[u_i w_{i,j}]$$

The first term is zero due to the residualization property:

$$E[\mathbf{X}_{i,(-j)} T (\beta_{(-j)} + \beta_j \alpha) w_{i,j}] = (\beta_{(-j)} + \beta_j \alpha) 1 E[\mathbf{X}_{i,(-j)} w_{i,j}] = 0$$

The third term is zero under the OLS ZCM assumption, $E[u_i | \mathbf{X}_i] = 0$.

$$E[u_i w_{i,j}] = 0$$

$$E[Y_i w_{i,j}] = E[\beta_j w_{i,j}^2]$$

$$E[Y_i w_{i,j}] = \beta_j E[w_{i,j}^2]$$

Finally,

$$\beta_j = \frac{E[Y_i w_{i,j}]}{E[w_{i,j}^2]}$$

Question 5.d. Prove the Frisch-Waugh theorem for the *sample* coefficients by showing that

$$\hat{\beta}_j = \frac{\sum_{i=1}^n Y_i \hat{w}_i}{\sum_{i=1}^n \hat{w}_i^2}$$

where $\hat{w}_i$ are the residuals from an auxiliary OLS regression of X_{ji} on all of the other X 's

$$X_{ji} = X'_{(\sim j)i} \hat{\alpha} + \hat{w}_i$$

PROOF: The full OLS model is $\mathbf{Y} = \mathbf{X}\hat{\beta} + \hat{\mathbf{u}}$. The OLS estimate for the j -th coefficient, $\hat{\beta}_j$.

$$\hat{\mathbf{w}}_j = \mathbf{X}_j - \mathbf{X}_{(-j)} \hat{\beta}_{(-j)}$$

The key property of OLS residuals is that they are **orthogonal** to the regressors:

$$\mathbf{X}'_{(-j)} \hat{\mathbf{w}}_j = \mathbf{0} \quad (*)$$

Next, The predicted value $\hat{\mathbf{Y}}$ from the full model can be written by partitioning the design matrix $\mathbf{X}$:

$$\mathbf{Y} = \mathbf{X}_{(-j)} \hat{\beta}_{(-j)} + \hat{\beta}_j \mathbf{X}_j + \hat{\mathbf{u}}$$

Substitute $\mathbf{X}_j = \hat{\mathbf{w}}_j + \mathbf{X}_{(-j)} \hat{\beta}_{(-j)}$ into the equation

$$\mathbf{Y} = \mathbf{X}_{(-j)} \hat{\beta}_{(-j)} + \hat{\beta}_j (\hat{\mathbf{w}}_j + \mathbf{X}_{(-j)} \hat{\beta}_{(-j)}) + \hat{\mathbf{u}}$$

$$\mathbf{Y} = \mathbf{X}_{(-j)} (\hat{\beta}_{(-j)} + \hat{\beta}_j \hat{\beta}_{(-j)}) + \hat{\beta}_j \hat{\mathbf{w}}_j + \hat{\mathbf{u}}$$

The normal equation for the coefficient $\hat{\beta}_j$ requires that the residuals $\hat{\mathbf{u}}$ are orthogonal to $\mathbf{X}_j$, which is:

$$\mathbf{X}'_j \hat{\mathbf{u}} = 0$$

$$\hat{\mathbf{w}}'_j \mathbf{Y} = \hat{\mathbf{w}}'_j \mathbf{X}_{(-j)} (\hat{\beta}_{(-j)} + \hat{\beta}_j \hat{\beta}_{(-j)}) + \hat{\mathbf{w}}'_j (\hat{\beta}_j \hat{\mathbf{w}}_j) + \hat{\mathbf{w}}'_j \hat{\mathbf{u}}$$

The first term is zero,

$$\hat{\mathbf{w}}'_j \mathbf{X}_{(-j)} (\dots) = 0$$

The last term is zero,

$$\hat{\mathbf{w}}'_j \hat{\mathbf{u}} = 0$$

$$\hat{\mathbf{w}}'_j \mathbf{Y} = \hat{\beta}_j (\hat{\mathbf{w}}'_j \hat{\mathbf{w}}_j)$$

$(\mathbf{A}'\mathbf{B} = \sum a_i b_i)$:

$$\sum_{i=1}^n Y_i \hat{w}_{i,j} = \hat{\beta}_j \sum_{i=1}^n \hat{w}_{i,j}^2$$

Solving for $\hat{\beta}_j$:

$$\hat{\beta}_j = \frac{\sum_{i=1}^n Y_i \hat{w}_{i,j}}{\sum_{i=1}^n \hat{w}_{i,j}^2}$$

1.6 Submission

Make sure you have run all cells in your notebook in order before exporting. **Please save before exporting.**